Marmara University Faculty of Engineering	**EE2002 FINAL EXAM**	**15.06.2023**

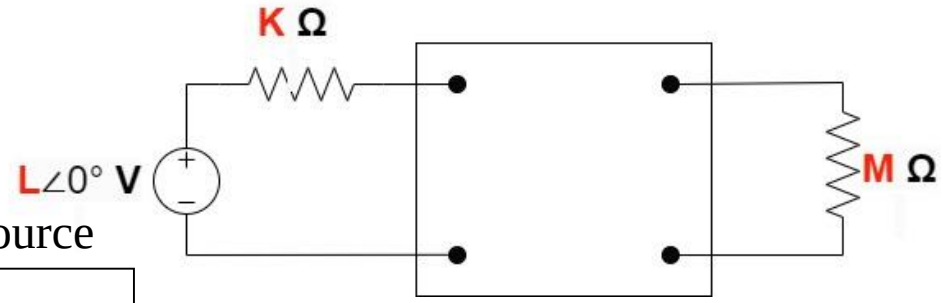
EXAM DURATION: **75 minutes**

NOTE: **In all the questions**, please consider **the last four digits of your student number** as **K, L, M** and **N** parameters. If **any number is 0**, use **the first non-zero number at the right side** of it. If your last number is 0, use **the first non-zero number at the left side** of it

Example: 150623045 → K=3, L=4, M=4, N=5

QUESTION-1 (35 pts): The **terminated two-port circuit** shown below is described in terms of its **h parameters** which are

$h_{11} = K$ $h_{12} = L$ $h_{21} = M$ $h_{22} = N$



a) (15 pts) Find the phasor voltage V_2.

b) (10 pts) Find the average power delivered to the load

c) (10 pts) Find the average power provided by the voltage source

The formulas you may need:

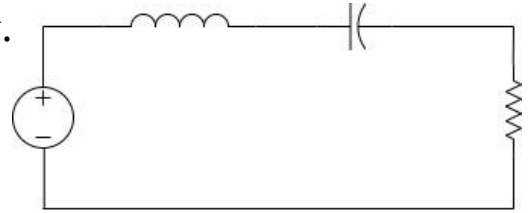
$$Z_{in} = h_{11} - \frac{h_{12}h_{21}Z_L}{1+h_{22}Z_L} \qquad \frac{V_2}{V_g} = \frac{-h_{21}Z_L}{(h_{11}+Z_g)(1+h_{22}Z_L)-h_{12}h_{21}Z_L}$$

QUESTION-2 (35 pts): Design a **bandpass passive filter** that allows signals within **N Hz and 20 Hz** frequency band and considering **M µF capacitor. Calculate,**

a) **(10 pts)** The center freuqency.

b) **(15 pts)** R and L

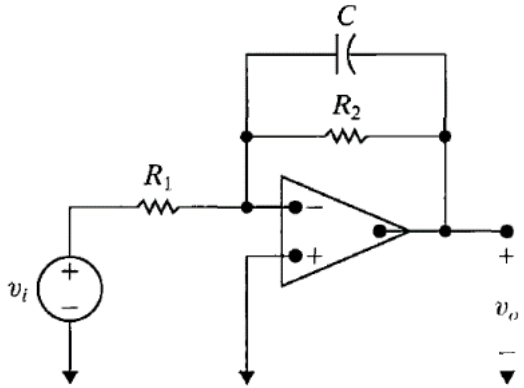
c) **(10 pts)** The quality factor.



The formulas you may need:

$$f_o = \sqrt{f_{c1}f_{c2}} \qquad \omega_o = \frac{1}{\sqrt{LC}} \qquad Q = \frac{f_o}{f_{c2}-f_{c1}} \qquad Q = \sqrt{\frac{L}{CR^2}}$$

QUESTION-3 (30 pts): Design a low-pass opamp active filter with a gain of **K,** cutoff frequency of **M Hz** and $R_1 = N\ \Omega$. **Calculate R_2 and C**



The formulas you may need:

$$K = \frac{R_2}{R_1}$$

$$\omega_c = \frac{1}{R_2 C}$$